

UNIT 4 : Semantic Parsing

❖ 4.1 Word Senses and WordNet

Word senses refer to the different meanings that a word can have in different contexts. In natural language processing (NLP), understanding and disambiguating word senses is crucial for tasks such as machine translation, information retrieval, and question answering.

WordNet is a lexical database commonly used in NLP to represent word senses and their relationships. It organizes words into sets of synonyms called "synsets," where each synset represents a distinct concept or meaning. WordNet includes a large collection of words, along with their definitions, usage examples, and relations to other words.

In WordNet, each synset is assigned a unique identifier and contains a list of words (lemmas) that share the same meaning. Synsets are connected through semantic relations, such as hypernymy (is-a relationship) and hyponymy (specific instance relationship). For example, the synset for "car" might include words like "automobile," "vehicle," and "motorcar," which are all hyponyms of the more general concept "car."

WordNet provides a hierarchical structure that allows for traversing from general to specific concepts. This structure can be useful in NLP tasks that require disambiguating word senses or finding related words. For instance, given an ambiguous word like "bank," WordNet can help determine if it refers to a financial institution or the side of a river by examining the relationships between synsets.

Several NLP libraries and frameworks provide access to WordNet, allowing developers to leverage its rich word sense information. These libraries often

provide methods for querying synsets, retrieving definitions, finding synonyms and antonyms, and exploring the hierarchical relationships between concepts.

Overall, WordNet plays a significant role in NLP by enabling the representation and exploration of word senses, contributing to tasks such as word sense disambiguation, semantic similarity calculation, and improving the understanding of language by machines.

❖ 4.1.1 Word senses

Consider the word "bank," which can have multiple senses depending on the context. Here are two distinct word senses for "bank":

Bank (noun): Financial institution

Example sentence: "I need to deposit my paycheck in the bank."

In this sense, "bank" refers to a financial institution where people deposit and withdraw money.

Bank (noun): Land alongside a body of water

Example sentence: "They sat on the bank of the river, enjoying the view."

Here, "bank" refers to the sloping land along the side of a river or any body of water.

In NLP, word sense disambiguation is the task of determining the correct sense of a word in a given context. By leveraging resources like WordNet or contextual information from the surrounding words, NLP models aim to assign the most appropriate sense to ambiguous words.

For example, if we have the sentence "I took out a loan from the bank," an NLP system needs to determine the correct sense of "bank" based on the context. By analyzing the sentence and considering the surrounding words, the system should infer that in this case, "bank" refers to a financial institution.

Word sense disambiguation is a challenging task, especially when the context is limited or ambiguous. However, it plays a crucial role in various NLP applications like machine translation, information retrieval, and sentiment analysis, where accurate understanding of word senses is essential for producing meaningful and accurate results.

4.1.2 Relation between senses

In NLP, the relation between senses is an important aspect for understanding and disambiguating word meanings. Here are some common relations between senses:

Synonymy:

Synonymy refers to the relation between two words or senses that have the same or very similar meanings. Identifying synonyms is useful for tasks like information retrieval, text summarization, and word sense disambiguation.

Antonymy:

Antonymy represents the relation between two words or senses that have opposite meanings. Recognizing antonyms can help in various NLP applications, such as sentiment analysis, question answering, and text generation.

Hypernymy/Hyponymy:

Hypernymy and hyponymy are hierarchical relations between concepts. A hypernym is a more general concept that encompasses one or more hyponyms, which are more specific instances. For example, "animal" is a hypernym of "cat"

and "dog." Understanding these relations is essential for tasks like information extraction, word sense disambiguation, and building knowledge graphs.

Meronymy/Holonymy:

Meronymy and holonymy describe the part-whole relationship between concepts. A meronym represents a part or component of a whole, while a holonym represents the whole to which the part belongs. For instance, "finger" is a meronym of "hand," and "hand" is a holonym of "finger." Recognizing meronym-holonym relations can aid in tasks such as text summarization, semantic role labeling, and text understanding.

Co-hyponymy:

Co-hyponymy refers to the relationship between two or more words or senses that share the same hypernym but are not synonyms or antonyms of each other. For example, "lion" and "tiger" are co-hyponyms of the hypernym "big cat." Understanding co-hyponymy is useful for tasks like information retrieval, text classification, and word sense disambiguation.

Polysemy:

Polysemy is the phenomenon where a single word has multiple related meanings. These different meanings are called senses. Recognizing and disambiguating polysemous words is crucial for many NLP tasks, such as machine translation, information retrieval, and sentiment analysis.

4.1.3 WordNet

WordNet is a widely used lexical database in natural language processing (NLP). It is a large semantic network that organizes words into sets of synonyms called "synsets" and captures the relationships between them. Here are some key aspects of WordNet's usage in NLP:

Word Sense Disambiguation:

WordNet is often employed in word sense disambiguation tasks, where the goal is to determine the correct sense of an ambiguous word in a given context. By leveraging WordNet's synsets and their relationships, NLP systems can analyze the context and select the most appropriate sense for a word, improving the accuracy of various NLP applications.

Synonym Expansion:

WordNet provides an extensive collection of synonyms for words through its synsets. NLP systems can utilize WordNet to expand the vocabulary of a text by including synonyms from relevant synsets. This can be useful in tasks like information retrieval, text summarization, and text generation, where a diverse set of words is needed.

Semantic Similarity:

WordNet's hierarchical structure and relationships between synsets enable the calculation of semantic similarity between words or sentences. NLP algorithms can leverage WordNet to compute similarity scores based on the distance between synsets in the hierarchy. This information is useful for tasks such as semantic search, document clustering, and recommendation systems.

Word Relationship Exploration:

WordNet allows NLP systems to explore various relationships between words, such as hypernymy (is-a relationship), hyponymy (specific instance relationship), meronymy (part-whole relationship), and more. These relationships can be utilized for tasks like information extraction, knowledge graph construction, and ontological reasoning.

Lexical Resources:

WordNet provides additional lexical information such as word definitions, usage examples, and word frequency statistics. NLP systems can access these resources to improve their understanding of word meanings and usage patterns, enhancing tasks like text comprehension, language modeling, and natural language understanding.

Many NLP libraries and frameworks provide interfaces or APIs to access WordNet's functionalities, making it easier for developers and researchers to incorporate WordNet into their NLP pipelines. By leveraging the rich semantic information captured in WordNet, NLP systems can achieve better performance in various language-related tasks.

4.1.4 wordsense disambituiton

Word sense disambiguation (WSD) is a crucial task in natural language processing (NLP) that involves determining the correct sense or meaning of an ambiguous word in a given context. Ambiguity arises when a word has multiple meanings or senses, and the goal of WSD is to select the most appropriate sense based on the surrounding context.

Here are some key aspects of word sense disambiguation in NLP:

Ambiguity:

Many words in natural language have multiple senses, and determining the intended sense is essential for accurate language understanding. For example, the word "bank" can refer to a financial institution or the side of a river. Resolving such ambiguities is crucial for tasks like machine translation, information retrieval, and question answering.

Context:

Context plays a vital role in disambiguating word senses. The meaning of a word can be influenced by its surrounding words, phrases, or the broader linguistic context. NLP algorithms leverage this contextual information to determine the sense that fits best in a given context. This can involve analyzing the neighboring words, syntactic relationships, or even leveraging broader discourse or document-level context.

Knowledge Resources:

WordNet, as mentioned earlier, is a valuable resource for word sense disambiguation. It provides semantic information, including synsets (sets of synonymous words) and their relationships. NLP systems can utilize WordNet's hierarchical structure and relations between synsets to disambiguate word senses. Other resources, such as ontologies, domain-specific lexicons, or corpora annotated with sense labels, can also aid in the disambiguation process.

Supervised and Unsupervised Methods:

Word sense disambiguation can be tackled using both supervised and unsupervised approaches. Supervised methods rely on labeled data, where examples are annotated with the correct sense. Machine learning algorithms, such as support vector machines (SVMs), decision trees, or neural networks, can be trained on such data to classify ambiguous words. Unsupervised methods, on the other hand, do not require labeled data and typically involve clustering or probabilistic models to group similar contexts together.

Sense Inventory:

The choice of sense inventory is an important consideration in word sense disambiguation. Different sense inventories, such as WordNet, FrameNet, or BabelNet, have their own advantages and limitations. The selection of an appropriate sense inventory depends on the specific NLP task and the available resources.

Word sense disambiguation remains an active research area in NLP, and various techniques, including supervised learning, unsupervised learning, and hybrid approaches, are continually being explored to improve the accuracy and robustness of disambiguation systems. Effective word sense disambiguation is crucial for improving the performance of many NLP applications, including machine translation, information retrieval, text summarization, and question answering.

4.1.5 WSD algorithm and task

Word sense disambiguation (WSD) algorithms aim to determine the correct sense of an ambiguous word in a given context. Several approaches have been developed to tackle this task in natural language processing (NLP). Here are some common WSD algorithms and techniques:

Supervised Machine Learning:

In supervised WSD, machine learning algorithms are trained on labeled data, where each example is annotated with the correct sense. Features extracted from the context, such as neighboring words, part-of-speech tags, syntactic dependencies, or word embeddings, are used to train classifiers, such as support vector machines (SVMs), decision trees, or neural networks. The trained models can then predict the sense of new instances.

Unsupervised Clustering:

Unsupervised WSD methods aim to group similar contexts together without relying on labeled data. These methods typically involve clustering algorithms, such as k-means, hierarchical clustering, or spectral clustering. Similar contexts, based on features like word co-occurrence or distributional similarity, are grouped into clusters, and each cluster is assigned a sense label. The sense of a new instance is determined based on its similarity to the contexts in each cluster.

Knowledge-based Methods:

Knowledge-based WSD approaches leverage external resources, such as lexical databases like WordNet or domain-specific ontologies. These resources provide information about word senses, their relationships, and definitions. Techniques like Lesk algorithm utilize the overlap between the context of the ambiguous word and the definitions of its candidate senses in a knowledge resource to disambiguate the word. Other methods involve exploring semantic networks, ontological reasoning, or using graph-based algorithms to exploit the hierarchical relations between word senses.

Semi-supervised and Active Learning:

In scenarios where labeled data is limited, semi-supervised learning techniques can be employed. These methods combine a small set of labeled examples with a larger set of unlabeled data during training. Active learning approaches iteratively select the most informative instances from the unlabeled data for annotation, improving the WSD model's performance with minimal labeling effort.

Ensemble Methods:

Ensemble approaches combine the predictions of multiple WSD models to improve accuracy. Different algorithms, such as supervised, unsupervised, or knowledge-based methods, are trained independently, and their outputs are combined using voting, weighting, or stacking techniques. Ensemble methods often lead to better overall performance by leveraging the strengths of individual algorithms.

Evaluation of WSD algorithms is typically done using annotated benchmark datasets, such as SemCor or Senseval, where the correct senses of words in various contexts are provided. Common evaluation metrics include precision, recall, and F1-score, measuring the algorithm's ability to correctly assign the intended sense.

WSD is an active research area in NLP, and researchers continuously explore new techniques, including deep learning models, hybrid approaches, and leveraging large pre-trained language models, to improve the accuracy and robustness of word sense disambiguation systems.

Word Sense Induction (WSI) and Semantic Role Labeling (SRL) are two important tasks in Natural Language Processing (NLP).

1. Word Sense Induction (WSI):

Word Sense Induction is the process of automatically identifying and grouping words that have similar meanings or senses based on their usage in a given context. WSI aims to cluster instances of a word into different groups or senses, without relying on pre-defined sense inventories.

2. Semantic Role Labeling (SRL):

Semantic Role Labeling is a task in NLP that involves identifying and classifying the semantic roles of words or phrases in a sentence, based on their relationships with a predicate or verb. The goal of SRL is to determine the roles played by different entities in a sentence, such as the agent, patient, instrument, or location. SRL helps in understanding the underlying meaning and structure of a sentence by capturing the relationships between different elements.

Semantic Roles in NLP.

Semantic roles in NLP refer to the grammatical and semantic relationships between different elements of a sentence and their associated roles or functions. It involves identifying the roles played by various words or phrases in relation to the predicate or verb in a sentence. Semantic role labeling is the task of automatically assigning these roles to different elements of a sentence.

Here are some common semantic roles:

1. **Agent:** The entity that performs the action or initiates the event described by the verb. For example, in the sentence "John eats an apple," "John" is the agent.
2. **Patient/Theme:** The entity that undergoes the action or is affected by it. In the sentence "John eats an apple," "apple" is the patient/theme.

3. Instrument: The means or tool used to perform the action. For example, in the sentence "John cuts the paper with scissors," "scissors" is the instrument.
4. Experiencer: The entity that experiences a particular sensation or state. In the sentence "She loves chocolate," "she" is the experiencer.
5. Goal: The entity toward which something moves or is directed. For example, in the sentence "He ran to the park," "park" is the goal.

Semantic role labeling aims to identify and classify these roles for each word or phrase in a sentence. It provides a deeper understanding of the relationships between different elements in a sentence, enabling applications such as information extraction, question answering, and machine translation.

Diathesis Alteration:

Diathesis alteration, or syntactic alternation, involves changing the sentence structure while preserving meaning.

It includes transforming sentences from active voice to passive voice and vice versa.

The alteration affects the roles and positions of grammatical constituents.

It is important for tasks like semantic role labeling and syntax parsing in NLP.

Causative and dative alternations are common types of diathesis alteration.

Causative alternation changes the causative relationship between subject and object.

Dative alternation involves transforming the indirect object to a prepositional phrase.

Understanding diathesis alteration is crucial for accurate representation of entity-action relationships.

It plays a role in machine translation systems for transforming sentences across languages.

Capturing diathesis alteration enhances NLP applications by accounting for sentence structure variations.

problems with thematic roles

Thematic roles, also known as semantic roles, in NLP can present a few challenges:

1. **Ambiguity:** Assigning thematic roles can be challenging due to word sense ambiguity. Words may have multiple meanings or senses, and determining the appropriate role can be difficult without considering the context.
2. **Subjectivity:** Thematic roles can be subjective and vary depending on the linguistic and cultural context. Different languages may have different conventions for assigning roles, leading to inconsistencies in interpretation.
6. **Limited training data:** Annotated training data for thematic role labeling is often limited and costly to create. The lack of large-scale, diverse, and high-quality datasets can hinder the development and performance of role labeling models.

Addressing these challenges requires continued research and development in NLP, including improvements in word sense disambiguation, cross-linguistic analysis, standardization of role inventories, robust parsing techniques, and the availability of comprehensive training data.

Lexicon for sentiment, affect and connotation:

1. **Sentiment:** The lexicon for sentiment consists of words or phrases that indicate positive, negative, or neutral sentiment. It includes sentiment-bearing terms like "happy," "sad," "good," "bad," and "neutral." These lexicons help in sentiment analysis tasks to determine the overall sentiment expressed in text.
2. **Affect:** The affect lexicon comprises words that represent different emotions or affective states. It includes terms like "joy," "anger," "fear," "surprise," and

"disgust." Affect lexicons assist in identifying and analyzing the emotional content or affective expressions in text.

3. Connotation: The connotation lexicon encompasses words or phrases that carry implied or associated meanings beyond their literal definitions. It includes words with positive or negative connotations, such as "brave," "deceptive," "innocent," or "manipulative." Connotation lexicons help in understanding the nuanced or evaluative aspects of language.

These lexicons are valuable resources for sentiment analysis, emotion detection, and connotation analysis in NLP. They provide pre-defined sets of words with assigned sentiment, affective, or connotative values, allowing NLP models to identify and interpret the subjective aspects of text. These lexicons can be built manually through human annotation or automatically through statistical methods and can be expanded or adapted based on specific application domains or languages.

Creating Affect Lexicons by Human Labeling:

Creating affect lexicons by human labeling involves the process of manually annotating words or phrases with their corresponding affective categories or emotional labels. In this approach, human annotators are presented with a set of words or textual instances and are asked to assign the appropriate affective or emotional labels to each word.

The process typically involves the following steps:

- 1. Annotator guidelines:** Clear guidelines are provided to annotators, explaining the desired affective categories or emotional labels and any specific criteria for assigning them. For example, annotators may be instructed to label words as "joy," "anger," "fear," "surprise," or "disgust."
- 2. Annotation task:** Annotators are presented with words or text instances one at a time and asked to assign the appropriate affective labels based on their subjective interpretation of the emotional content conveyed by the word.
- 3. Multiple annotators:** To ensure reliability and reduce bias, multiple annotators are involved in the labeling process. Inter-annotator agreement measures, such as

Cohen's kappa, may be calculated to assess the level of agreement among annotators.

Creating affect lexicons through human labeling is a time-consuming and resource-intensive task. However, it allows for capturing the nuances and subtleties of affective expressions in language by leveraging human judgment and expertise. These annotated affect lexicons serve as valuable resources for training affect detection models and conducting research in affective computing and sentiment analysis.

Semi-supervised Induction of Affect Lexicons:

Semi-supervised induction of affect lexicons refers to a method that combines both labeled and unlabeled data to create lexicons containing words or phrases annotated with affective or emotional labels. In this approach, a limited set of words is initially labeled by human annotators, while a larger set of unlabeled words is used to supplement the labeled data.

The process typically involves the following steps:

1. Initial labeled data: A small set of words or phrases is manually annotated with affective labels by human annotators. These labeled instances serve as the initial seed data.
2. Feature extraction: A set of relevant features or characteristics is extracted from the labeled data and the unlabeled data. These features may include lexical, syntactic, or semantic properties associated with affect or emotion.
3. Training model: A machine learning model, such as a classifier or clustering algorithm, is trained using the labeled data and the extracted features. The model learns to associate the features with the affective labels.

Semi-supervised induction of affect lexicons leverages the unlabeled data to expand the coverage of affective terms beyond the initial labeled set. By utilizing the larger unlabeled dataset, the method can capture a wider range of affective expressions present in the language. This approach helps in reducing the reliance on manual labeling efforts while still producing useful and comprehensive affect lexicons for various affect analysis tasks in natural language processing.

Supervised Learning of Word Sentiment:

Supervised learning of word sentiment involves training a machine learning model to predict the sentiment or polarity of words based on labeled training data. In this approach, a dataset of words or phrases with their corresponding sentiment labels (e.g., positive, negative, neutral) is used to teach the model to recognize and classify sentiment.

The process typically involves the following steps:

1. Labeled training data: A collection of words or phrases is manually annotated with sentiment labels, indicating their positive, negative, or neutral sentiment. This labeled data serves as the training set.
2. Feature extraction: Relevant features are extracted from the labeled data to represent the words or phrases. These features can include lexical, syntactic, or contextual information, such as word frequency, part-of-speech tags, or neighboring words.
3. Model training: A supervised learning algorithm, such as logistic regression, support vector machines, or neural networks, is trained using the labeled data and the extracted features. The model learns to associate the features with the sentiment labels.

Supervised learning of word sentiment allows the model to learn from labeled data and generalize its knowledge to classify sentiment in previously unseen words. By leveraging machine learning algorithms and feature representations, this approach automates the sentiment analysis process and can be applied to various NLP tasks, such as sentiment classification, opinion mining, or sentiment lexicon expansion.

Personality, Affect Recognition, Lexicon-based methods for Entity-Centric Affect, Connotation Frames:

1. **Personality:** Personality refers to the unique set of characteristics, traits, and patterns of behavior that define an individual. In the context of NLP, personality analysis involves determining and predicting an individual's personality traits based on their language usage. Machine learning techniques and linguistic features are used to analyze text and extract information related to personality dimensions such as extraversion, agreeableness, openness, conscientiousness, and neuroticism.

2. **Affect Recognition:** Affect recognition focuses on the identification and understanding of emotions, moods, and sentiments expressed in text or speech. Machine learning algorithms, including deep learning approaches such as recurrent neural networks or transformers, are trained to classify or predict affective states based on linguistic features, contextual information, or acoustic cues in the case of speech. Affect recognition enables the automatic detection and analysis of emotional content in various applications, including sentiment analysis, emotion detection, and affective computing.

3. **Lexicon-based methods for Entity-Centric Affect:** Lexicon-based methods involve using pre-built lexicons or dictionaries containing words or phrases annotated with affective or emotional labels to determine the affect or emotion associated with entities in text. These lexicons assign sentiment or emotional values to specific words and are used to calculate the overall affective orientation or emotional intensity of entities mentioned in the text. Lexicon-based approaches provide a quick and computationally efficient way to analyze affect at the entity level, such as identifying the sentiment associated with specific named entities or entities in context.

4. **Connotation Frames:** Connotation frames represent the underlying evaluative associations and biases associated with words or concepts beyond their literal meanings. They capture the connotative or evaluative aspects of language, including positive, negative, or neutral connotations. Connotation frames provide a structured representation of how words or concepts are typically perceived or evaluated. They can be created through manual annotation or automatic methods and serve as valuable resources for sentiment analysis, connotation analysis, or understanding the evaluative nature of language.

These concepts and techniques play important roles in various NLP applications, ranging from sentiment analysis and affective computing to understanding the subjective aspects of language and evaluating the connotative associations of words or concepts.